

Simplex Algorithm

Goal is to find an optimal solution starting at a basic feasible solution

Def Simplex Multipliers mrt B

$$\pi = (B^{-1})^T c_b$$

Def Reduced Costs mrt B

$$c^\pi = c - A^T \pi$$

Note that $c^\pi = (c_B, c_L) - (B^T, L^T)\pi$

$$\rightarrow (c_B, c_L) - (B^T, L^T)(B^{-1})^T c_B$$

$$\rightarrow (c_B - B^T(B^{-1})^T c_B, c_L - L^T(B^{-1})^T c_B)$$

$$\rightarrow (0, c_L - L^T(B^{-1})^T c_B) \text{ (Note that } B^T(B^{-1})^T = (B^{-1}B)^T = I)$$

For finding equivalent points, $(c^\pi)^T x = c^T x - \pi^T A x = c^T x - \pi^T b$ (note $\pi^T b$ is a constant)

$$\text{This is because } \min c^T x = (c^\pi)^T x \equiv \min(c^\pi)^T x$$

Main Idea:

$$c_j^\pi < 0 \rightarrow \text{increase } x_j, \text{ update } x_B \text{ and repeat until optimal solution (no } c_j^\pi < 0)$$

Additionally note that you aim to increase the x_j with the largest increase to the objective (decrease in case of minimizing)

Small issue is that a change of basis doesn't necessarily change extreme points